Linear algebra – 1

Determinants and invertibility

Homework

12 сентября 2026 года

Theoretical minimum. The adjugate satisfies $A \operatorname{adj} A = (\det A)E$ and has rank n, 1 or 0 according to the rank of A; Jacobi's formula differentiates a determinant and gives $\det e^A = e^{\operatorname{tr} A}$. A determinant of a matrix which is a small-rank perturbation of a simple one is computed by $\det(A + UV^{\mathsf{T}}) = \det A \cdot \det(E + V^{\mathsf{T}} A^{-1} U)$. An integer matrix is invertible over $\mathbb{Z}$ exactly when $\det A = \pm 1$, and since det is a polynomial with integer coefficients, identities transfer to any commutative ring and reduction modulo a prime proves non-vanishing. Singularity has three equivalent faces — dependent columns, a zero eigenvalue, a non-trivial kernel — and the art is to pick the right one. Finally, over $\mathbb{R}$ a determinant is a continuous function of the entries, so a determinant identity proved on a dense set holds everywhere: this is how one passes from invertible matrices to all of them. A skew-symmetric matrix of odd order is singular, and a real skew-symmetric matrix has even rank. For a skew-symmetric A of order $2n$ the *Pfaffian*

$$\operatorname{Pf} A = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \operatorname{sgn} \sigma \prod_{i=1}^{n} a_{\sigma(2i-1)\sigma(2i)}$$

is a polynomial in the entries with $\det A = (\operatorname{Pf} A)^2$; it expands along the first row as

$$\operatorname{Pf} A = \sum_{j=2}^{2n} (-1)^j a_{1j} \operatorname{Pf} A_{\hat{1}\hat{j}},$$

where $A_{\hat{1}\hat{j}}$ is A without the rows and columns 1 and j.

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1. (VJIMC 2013) Let $A = (a_{ij})$ and $B = (b_{ij})$ be two real 10×10 matrices such that $a_{ij} = b_{ij} + 1$ for all i, j and $A^3 = O$. Prove that $\det B = 0$.

2. (IMC 2015) For any integer $n \geqslant 2$ and two $n \times n$ matrices with real entries A, B satisfying $A^{-1} + B^{-1} = (A + B)^{-1}$, prove that $\det A = \det B$. Does the same conclusion hold for matrices with complex entries?

3. (IMC 2014) Determine all pairs (a, b) of real numbers for which there exists a unique symmetric 2×2 matrix M with real entries satisfying $\operatorname{tr} M = a$ and $\det M = b$.

4. (Romania national olympiad 2017) Let $n \geqslant 2$ and k be natural numbers and let $A_1, \ldots, A_k$ be symmetric real $n \times n$ matrices. Prove that the following are equivalent:

   1) $\det\left(\sum_{i=1}^{k} A_i^2\right) = 0$;

   2) $\det\left(\sum_{i=1}^{k} A_i B_i\right) = 0$ for all $B_1, \ldots, B_k \in M_n(\mathbb{R})$.

5. (Romania national olympiad 1995) Let n, k be positive integers and let $A_i \in M_n(\mathbb{R})$, $i = 1, \ldots, k$. Prove that $\det\left(\sum_{i=1}^{k} A_i^{\mathsf{T}} A_i\right) \geqslant 0$.

6. (Romania national olympiad 2011) Let A be a singular real matrix of order $n \geqslant 2$ and let $\operatorname{adj} A$ be its adjugate. Prove that $\operatorname{tr}(\operatorname{adj} A) \neq -1$ if and only if the matrix $E + \operatorname{adj} A$ is invertible.

7. (Romania national olympiad 2008) Let A be an $n \times n$ matrix with complex entries. Prove that $A^{-1} = \overline{A}$ if and only if there exists an invertible complex matrix B with $A = B^{-1}\overline{B}$.

8. (Romania national olympiad 2025) Let n be a positive integer and let a, b be complex numbers with $a \neq 0$ and $b^k \neq 1$ for every $k \in \{1, 2, \ldots, n\}$. The matrices $A, B \in M_n(\mathbb{C})$ satisfy $BA = aE + bAB$. Prove that A and B are invertible.

9. (Romania national olympiad 2012) Let $n \geqslant 2$ and $1 \leqslant k \leqslant n-1$. Prove that if $A \in M_n(\mathbb{C})$ has exactly k minors of order $n - 1$ equal to zero, then $\det A \neq 0$.

10. (Romania national olympiad 2014) Let $A \in M_4(\mathbb{R})$ be invertible with $\operatorname{tr} A = \operatorname{tr}(\operatorname{adj} A) \neq 0$. Show that $A^2 + E$ is singular if and only if there is a non-zero matrix in $M_4(\mathbb{R})$ anticommuting with A.

11. (VJIMC 1991) Compute the determinant of an antisymmetric matrix $(a_{ij})_{i,j=1}^{n}$, where $a_{ij} = -a_{ji}$ and $a_{ii} = 0$, for odd n, and explain what changes for even n.

12. (Мехмат, 2007) Prove Carleman's inequality: for every real matrix $A = (a_{ij})_{i,j\leqslant n}$

$$|\det(E + A)| \leqslant \exp\Big(\operatorname{tr} A + \tfrac{1}{2}\sum_{i,j=1}^{n} a_{ij}^2\Big).$$

13. Prove that if all entries of an integer $n \times n$ matrix are odd, then 2^{n-1} divides its determinant, and show that the exponent $n-1$ cannot be improved.

14. (Putnam 2008) Alan and Barbara play a game in which they take turns filling entries of an initially empty 2008×2008 array. Alan plays first. At each turn a player chooses a real number and places it in a vacant entry. The game ends when all the entries are filled. Alan wins if the determinant of the resulting matrix is non-zero, Barbara wins if it is zero. Which player has a winning strategy?

15. (Putnam 2002) In Determinant Tic-Tac-Toe, Player 1 enters a 1 in an empty 3×3 matrix, Player 0 counters with a 0 in a vacant position, and play continues in turn until the matrix is filled with five ones and four zeros. Player 0 wins if the determinant is 0 and Player 1 wins otherwise. Assuming optimal play, who wins and how?

16. (CIIM 2009) Determine whether for every natural n there is a real $n \times n$ matrix whose determinant is zero but such that changing any single entry produces a matrix with non-zero determinant.